



3D map and mmWave radar-based self-localization for UAVs in GNSS-denied environments

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ABSTRACT

Reliable self-localization of unmanned aerial vehicles (UAVs) in dense urban environments remains a major challenge due to the frequent unavailability or degradation of Global Navigation Satellite Systems (GNSS) and other radio signals. This paper presents a robust and cost-effective method for UAV self-localization by using vision and millimeter-wave (mmWave) radar data in GNSS-denied environments. The approach generates an initial dense point cloud through depth estimation and semantic segmentation, which is then geometrically refined using sparse mmWave radar point cloud. A semantic-guided clustering method is applied to the mmWave radar point cloud to remove noise and extract key structural elements such as walls, which are later fused with vision-based depth information. For positioning, image matching algorithm provides coarse localization, followed by fine registration that leverages geometric features of windows to enhance precision. Experimental results demonstrate that the proposed method can achieve self-localization accuracy within 0.4 m, while maintaining low system complexity and deployment cost, offering a practical solution for UAV self-localization in GNSS-denied urban scenarios.

1. Introduction

Driven by advancements in autonomous flight and sensing technologies, the unmanned aerial vehicles (UAVs) have found widespread use in diverse applications such as surveying [1], power line inspection [2], autonomous delivery [3], regional monitoring [4], and emergency search and rescue [5], playing an increasingly important role in both industrial and civilian domains. Accurate navigation and positioning capabilities have become a fundamental prerequisite for the reliable execution of drone operations. Currently, most drone systems rely on a combination of Global Navigation Satellite Systems (GNSS) and Inertial Navigation Systems (INS), where GNSS corrects accumulated drift of inertial sensors to enable high-precision self-localization. Nevertheless, in environments with weak or absent GNSS signals, including urban canyons, indoor areas, and complex terrains, the performance of the integrated navigation system declines markedly, critically affecting the drone's autonomy [6–8]. Therefore, achieving robust self-localization and navigation in GNSS-denied environments remains a significant challenge with growing research interest.

With the growing demand for UAV deployment in complex environments, reliable self-localization is a fundamental requirement for

autonomous operation. This requires not only building accurate 3D environmental models but also aligning them with prior maps to enable effective spatial registration. In such contexts, 3D environmental perception serves as the foundation for understanding the surroundings, providing raw spatial information such as geometry and semantics. Building on this foundation, environmental modeling plays a central role in transforming raw observations into structured representations, facilitating map alignment and self-localization. Ultimately, self-localization acts as the vital bridge connecting perception with subsequent tasks such as emergency search and autonomous delivery, as presented in Fig. 1. Hence, developing an accurate and robust 3D perception system is thus essential for reliable environmental understanding and high-precision UAV self-localization.

1.1. Related work

Various positioning technologies have been developed and applied to achieve accurate drone self-localization. Among them, vision-based methods are widely adopted due to their high accuracy, low cost, and ease of implementation. These methods typically rely on monocular or stereo cameras to capture images of the environment, from which

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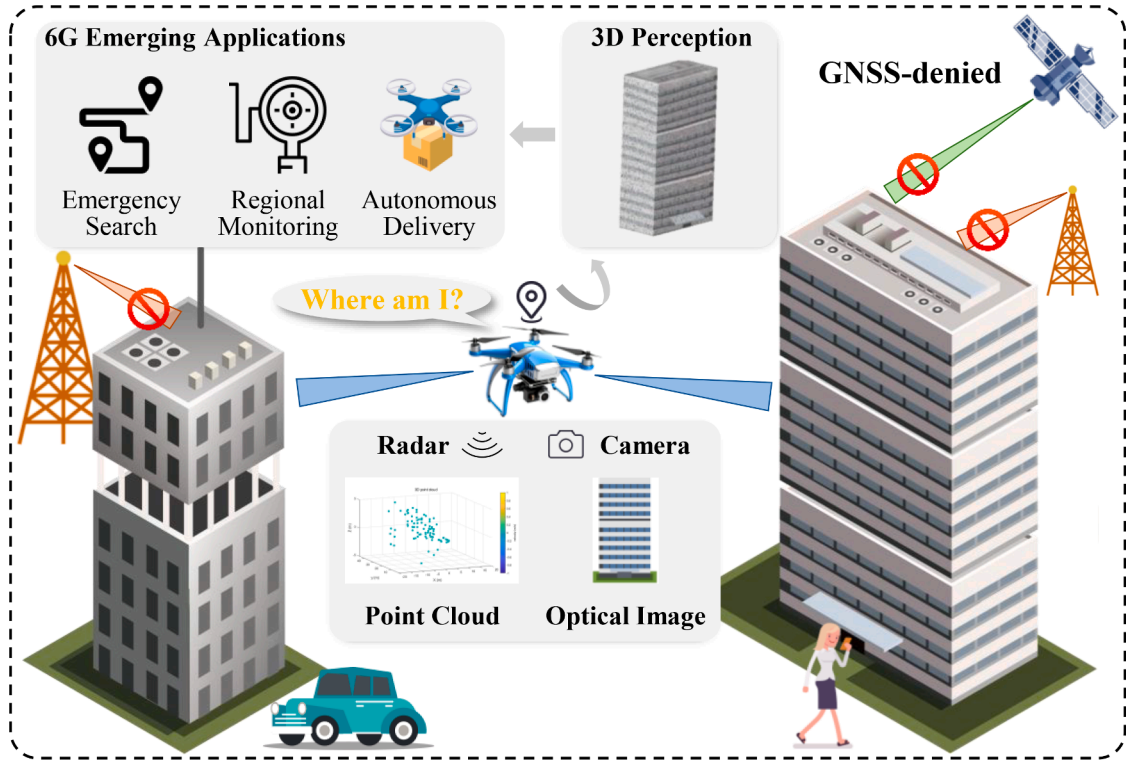


Fig. 1. An overview of self-localization for UAVs in GNSS-denied environments.

salient visual features (e.g., Oriented FAST and Rotated BRIEF [9], Scale-Invariant Feature Transform (SIFT) [10], or learned descriptors [11]) are extracted and matched across frames. By analyzing the motion of these features over time or against a pre-built map, the system estimates the camera's relative pose using Structure-from-Motion (SfM), Visual Odometry (VO), or Visual SLAM techniques [12]. Furthermore, deep learning-based approaches have been introduced to enhance robustness under dynamic conditions by learning more discriminative and invariant representations. Qiu et al. [13] proposed a framework that models the self-localization process as a combination of image retrieval, feature matching, and geometric transformation (e.g., Perspective-n-Point (PnP)), enhanced by resolution adaptation and feature quality selection modules to improve projection consistency and matching accuracy. Liu et al. [14] introduced a multisource feature learning framework that align aerial images with georeferenced satellite images using Siamese networks and spatial attention, enabling robust localization even under heterogeneous imaging conditions. Visual-based localization methods have shown impressive performance in environments with good visibility. However, their performance degrades significantly in environments with poor lighting or low-texture surfaces, where the perception capability of visual sensors is severely limited. Moreover, the reliance on complex feature extraction and matching pipelines often results in high computational complexity and increased energy consumption, which restricts their deployment in real-time UAV applications. These limitations have motivated the exploration of alternative sensing modalities that can provide more robust geometric information.

Owing to high range accuracy and direct acquisition of geometric information, positioning methods based on Light Detection and Ranging (LiDAR) technology is the much-sought techniques in this area. Qian et al. [15] employed LiDAR and monocular cameras to construct textured 3D point cloud maps, leveraging LiDAR odometry to provide prior pose and depth information, which was combined with visual matching to optimize pose estimation and enable robust, real-time localization. Petráček et al. [16] utilized readily available 3D point clouds and 2D LiDAR to build local maps, integrating onboard Inertial

Measurement Unit (IMU) data to achieve accurate positioning in dark areas of large historical sites. Xu et al. [17] proposed an asynchronous LiDAR–camera fusion framework to estimate the absolute pose and position of drones in unknown outdoor environments. Despite their effectiveness, LiDAR systems are often constrained by high power demands, substantial weight, and sensitivity to adverse weather conditions, which limit their broader deployment in drone applications. Given the limitations in power, weight, and environmental robustness of LiDAR systems, lightweight radio-based techniques such as UWB are increasingly explored as efficient alternatives for UAV localization.

Among such alternatives, ultrasonic [18] and ultra-wideband (UWB) [19] positioning technologies share similar operating principles, both relying on pre-deployed auxiliary nodes for localization. While ultrasonic systems can achieve centimeter-level accuracy, their low update rate and susceptibility to occlusion will result in the instability of the positioning accuracy [20]. UWB systems leverage systems to large bandwidth and nanosecond pulses to deliver high temporal resolution and precise positioning [21,22], but their performance in complex environments usually requires assistance from other sensors such as GPS, IMU, or vision to maintain robustness. Such dependence reflects a broader challenge in UAV self-localization, where most existing solutions continue to rely heavily on external infrastructure.

Even with significant progress in improving the accuracy of self-localization systems for UAVs, most existing solutions continue to depend heavily on external infrastructure, which leads to high deployment and maintenance costs. In addition, mainstream perception methods such as LiDAR, INS, and vision present inherent limitations, making it difficult to maintain stable performance in complex and dynamic environments.

1.2. Motivation

The increasing adoption of real-world 3D models has rendered them a valuable resource for enhancing the robustness of self-localization, as they provide rich geometric structures and semantic information

for prior map support [23,24]. Meanwhile, millimeter-wave (mmWave) radar has been introduced to further extend environmental perception capabilities [25], offering strong penetration capabilities and reliable performance under poor lighting and adverse weather conditions, allowing the capture of structural information even in visually degraded environments [26,27]. Nevertheless, the generated point cloud tends to be sparse and insufficiently detailed, limiting its suitability for high-quality 3D reconstruction on its own. Advances in visual perception techniques, particularly in depth estimation and semantic segmentation [28–30], have enabled the extraction of spatial and semantic information from single images, laying a robust foundation for multimodal fusion. Depth prediction models exhibit strong generalization ability and can generate dense depth maps from unlabeled data, while segmentation models such as Mask2Former demonstrate excellent transferability across datasets [31], facilitating the automatic identification of semantic regions and categories without manual annotation. The combination of these visual modalities forms a solid basis for recovering 3D structure from monocular imagery. The fusion of visual and radar sensing technologies combines the strong generalization capability of visual segmentation models with the robustness of radar against adverse environmental conditions such as poor lighting and weather, thereby supporting 3D scene reconstruction and precise self-localization across diverse environments.

Despite the complementary strengths of mmWave radar and monocular cameras, existing research primarily focuses on sensor-level fusion for object detection or depth completion, with few studies addressing their joint potential for large-scale 3D reconstruction and self-localization. Most existing 3D mapping and localization methods, especially for large-scale scenarios, still rely on LiDAR sensors. Unlike these LiDAR-based approaches, where dense and geometrically uniform point clouds facilitate direct geometric processing, the combination of mmWave radar and monocular cameras introduces significant heterogeneity at the data level. Specifically, mmWave radar provides sparse and noisy 3D point measurements, lacking appearance information, while monocular images contain rich texture and semantic cues but without direct depth perception. These two modalities differ fundamentally in terms of sensing mechanism, spatial resolution, and data representation, forming a typical heterogeneous sensing problem.

Furthermore, the prior real-scene 3D model used for localization is typically represented as a dense point cloud or textured mesh, adding another modality gap between local observations and the global map. Bridging this multi-level heterogeneity, from raw sensor data fusion to observation-to-map matching, remains a key technical challenge for enabling accurate and robust 3D reconstruction and self-localization in GNSS-denied environments. The sparse and noisy nature of radar point clouds poses significant challenges in structural feature extraction and precise map matching, which are essential for accurate self-localization within real-scene 3D model.

1.3. Contribution

This paper proposes a novel approach for 3D mapping and precise self-localization of UAVs in GNSS-denied environments based on the fusion of mmWave radar and visual information. A dense point cloud is generated from monocular images via depth inference and semantic segmentation, while geometric measurements from mmWave radar are integrated to improve structural completeness and spatial accuracy without relying on costly equipment or extensive labeled datasets. The resulting local point cloud is then matched with high-precision real-world 3D maps to facilitate robust UAV self-localization in complex urban environments with window structures serving as geometric anchors for fine matching after initial image-based alignment.

The main contributions of this paper are given as follows:

- 1) We propose a mmWave radar-based method for 3D mapping and self-localization that integrates visual information and mmWave radar data, combining semantic understanding from images with geomet-

Table 1

Parameters and their descriptions.

Parameters	Their descriptions
p_i/p_j	Query point in the point cloud and its neighbor candidate point
$\mathcal{N}(p_i, r)$	Neighborhood set of p_i within radius r
τ	Minimum number of points threshold
D	Point cloud dataset
d_i	Distance from a point to the line
$\hat{d}_1/\hat{d}_2$	Distance threshold for inliers/line duplication
v_1, v_2	Unit direction vectors of two lines with angle θ
L_{rad}, L_{rec}	Structural lines from radar and reconstructed point clouds
$D_r, D_{r \rightarrow img}, D_{recon}$	Radar point set, projected radar points, reconstructed points
λ	Similarity threshold for structural line correspondence
s	Scaling factor between radar and reconstructed points
x_i	3D point coordinate in point cloud with centroid c
α/β	Weight coefficient of geometric fitting/void region alignment
$\tilde{p}_i/\tilde{p}'_i$	Point/Its corresponding point in source/target point cloud
$\tilde{b}_k/\tilde{b}'_k$	Boundary points of the window area/Its corresponding point in reconstructed/real-scene point cloud
v/w	Pixel/Radar coordinate
M	Camera intrinsic matrix
Z_c	Depth (Z-axis) in camera coordinates
x'_p, y'_p	Projected pixel coordinates from radar point
N_c	Number of calibration correspondences
N_o	Number of observations in i th experiment
$\Delta x, \Delta y, \Delta z$	Differences between estimated and reference UAV position along X, Y, Z-axes
e_{geo}, e_{void}	Geometric and void-region alignment errors
e_{proj}	Average calibration error between projected radar points and pixel points
e_{loc}	Drone positioning error

ric correction from radar. By constraining image-based depth estimation using radar point clouds, the method significantly enhances reconstruction accuracy and self-localization performance in GNSS-denied environments.

- 2) We introduce a semantic-guided approach for point cloud clustering and structural extraction. Semantic masks derived from images are combined with density constraints to eliminate non-architectural regions and outliers. An improved random sample consensus (RANSAC) algorithm then extracts wall structures and establishes geometric correspondences, providing robust priors for matching. Experimental results illustrate our method yields over 90 % clustering precision, focusing on critical geometric components.
- 3) We present a window-assisted registration method based on local geometric constraints. Built upon coarse image matching, semantic anchors are constructed from window regions by integrating color gradients, spatial positions, and contour features, serving as localized constraints to guide precise point cloud alignment. Experimental verification shows that our method achieves self-localization errors below 0.4 m.

The rest of the paper is organized as follows. Section 2 introduces the setup overview and preprocessing of real scene 3D model data. Section 3 describes the proposed self-localization algorithm based on the mmWave radar and camera. Section 4 presents the experimental results in a typical scene. Section 5 concludes this paper.

2. Setup overview and preprocessing of real-scene 3D model data

In this section, we present the system configuration for UAV self-localization in GNSS-denied environments and the construction of a paired 3D point cloud and 2D image dataset derived from segmented regions of a real-scene 3D model. This paired datasets provide aligned geometric and visual information, enabling effective sensor fusion and accurate observation-to-map matching for robust UAV localization. For clarification, we summarize the important parameters with their descriptions in Table 1.

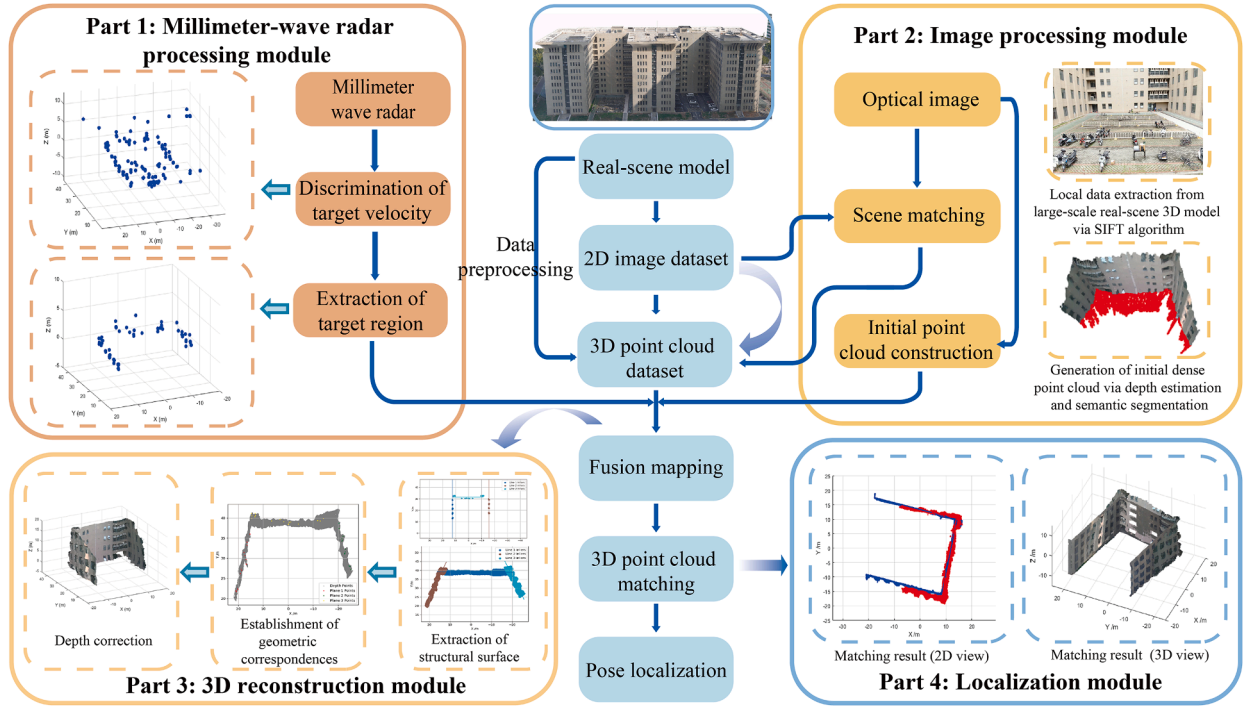


Fig. 2. Block diagram of the self-localization method.

2.1. Setup overview

Fig. 1 illustrates an overview of the UAV self-localization system in GNSS-denied environment. In GNSS-denied environments, a stationary unmanned platform is equipped with a mmWave radar and a monocular camera, enabling 3D mapping of surrounding structures as well as self-localization within a prior 3D model. This sensor configuration exploits the complementary advantages of mmWave radar and monocular vision, combining robustness of radar in adverse weather and low visibility with camera-based texture and color perception, facilitating both structural reconstruction and pose estimation relative to the environment. In particular, the localization approach benefits from wide field-of-view depth sensing of radar combined with visual feature extraction, enhancing robustness in complex environments¹.

The structural landmarks utilized in our method are planar wall surfaces and corner regions, which satisfy the requirements of being ubiquitous, structurally distinct, and extractable from sparse radar point clouds. These landmarks are represented as simplified geometric primitives (planes and edges), facilitating efficient matching against the prior real-scene 3D model of the environment. Due to the non-unique nature of these geometric features, accurate localization requires either a coarse prior pose to resolve ambiguities. We will demonstrate that even under sparse radar data conditions, the extracted structural features are sufficient to enable accurate self-localization within the 3D real-scene model. Fig. 2 outlines the overall processing pipeline followed throughout this paper.

2.2. Preprocessing of real-scene 3D model data

Real-scene 3D models are accurate digital reconstructions generated from multi-angle drone imagery using photogrammetry and 3D reconstruction techniques, offering centimeter-level precision, complete spa-

tial structures, photorealistic textures, and accurate geographic coordinates.

Considering the large volume and redundancy of complete real-world 3D model data, which poses challenges for validating autonomous UAV positioning algorithms, this paper segments the original model to construct corresponding image and point cloud datasets, as shown in Fig. 3. During the segmentation process, the area is divided using a fixed voxel size of $30\text{ m} \times 30\text{ m} \times 10\text{ m}$, with adjacent segments overlapping at 1-meter intervals. The cutting data are named in a fixed format: for buildings with numerical labels, the format is “numerical label + building + annotation + slice number”; for unlabeled areas, the format is “place name + annotation + slice number”. After segmenting the 3D data, corresponding 2D images are generated using the same naming convention as the point cloud dataset, ensuring a one-to-one correspondence that facilitates subsequent data processing and analysis.

3. Proposed method

This section details the proposed self-localization algorithm for UAV spatial localization, as depicted in Fig. 2. The process starts with preprocessing images and radar point clouds, with detection results from both sources as input. Visual 3D points are then geometrically refined using mmWave radar to achieve effective data fusion. Subsequently, optical image matching is performed using the SIFT algorithm with the prepared 2D image dataset, retrieving the corresponding real-scene 3D model segments for coarse localization. Finally, pose estimation is conducted by aligning with the real-scene point cloud model, resulting in accurate self-localization.

3.1. Processing of multi-modal data and preliminary point cloud generation

In outdoor 3D perception and positioning tasks for unmanned systems, single sensor approaches often face trade-offs among accuracy, robustness, and structural completeness. This paper proposes a mmWave-based strategy that integrates vision to exploit complementary strengths in geometry and stability, enabling more reliable environmental perception.

¹ While this paper demonstrated with a single mmWave radar and monocular camera, the proposed methods readily extend to stereo vision or multi-radar configurations.

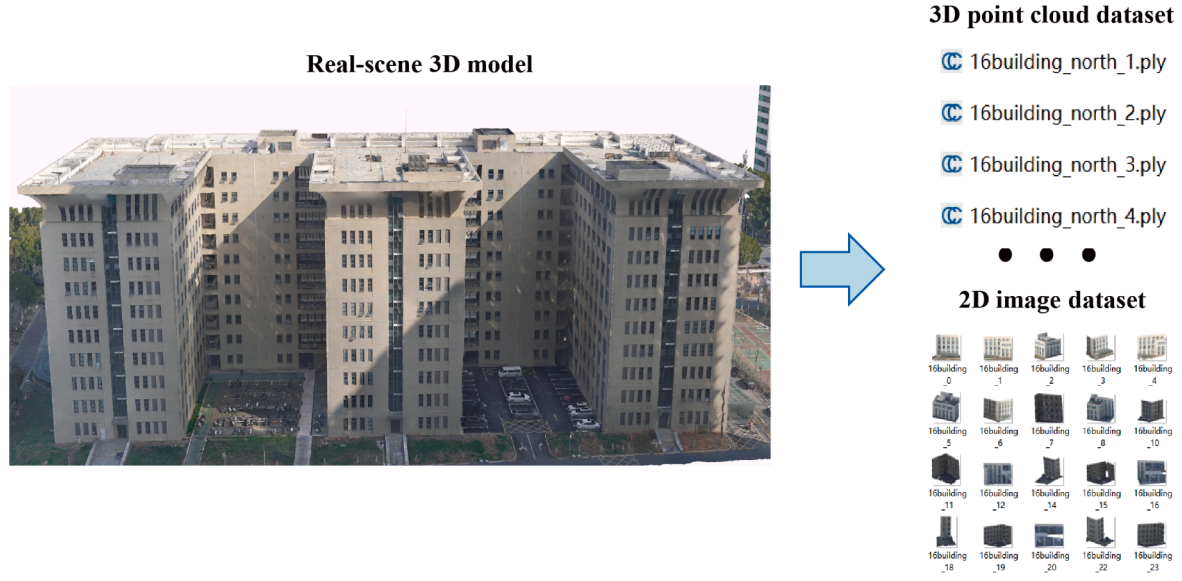


Fig. 3. Building a corresponding 3D point cloud and 2D image dataset from real-scene 3D models.



Fig. 4. Semantic segmentation results. (a) Inference result from the Mask2Former model. (b) Extracted building mask.

3.1.1. Image information extraction

This paper utilizes existing deep learning models to extract two key types of information from images: depth and semantic category data. With advances in CNN- and Transformer-based approaches, semantic segmentation and depth estimation have evolved into end-to-end, data-driven tasks, significantly improving model expressiveness and robustness.

Semantic segmentation aims to assign a category label to each pixel for fine-grained scene understanding. Mask2Former leverages the powerful capabilities of the Transformer architecture to build a unified and efficient model that adapts to various segmentation tasks via mask-based learning, achieving strong generalization and fully automatic image segmentation without user prompts. As shown in Fig. 4, we adopt the publicly available Mask2Former model trained on the COCO dataset to obtain building-related semantic masks and corresponding category labels.

Monocular depth estimation predicts per-pixel depth from a single image, supporting the recovery of the overall 3D structure. Depth Anything V2 offers robust monocular depth inference with strong zero-shot transferability, generating dense and continuous depth maps from a single image for 3D reconstruction in complex environments. An initial dense point cloud is generated by integrating semantic segmentation masks with monocular depth information. To address frequent depth discontinuities at object boundaries, point cloud normal vectors are utilized to identify and eliminate edge anomalies, enhancing geometric consistency and supporting more reliable downstream processing, as depicted in Fig. 5.

3.1.2. Radar information extraction

The mmWave radar provides accurate ranging and strong resistance to interference under diverse weather conditions, but its sparse point clouds with limited structural features hinder reliable point cloud clustering in complex environments.

We propose a semantic-guided clustering method that projects radar point clouds onto the image plane and retains only the points falling within building regions identified by semantic segmentation. A density constraint is then applied to remove isolated noise. For any query point $p_i \in \mathbb{R}^3$ in the point cloud, the set of p_i 's neighborhood is defined as:

$$\mathcal{N}(p_i, r) = \{p_j \in \mathbb{R}^3 \mid \|p_j - p_i\|_2 \leq r, j \neq i\}, \quad (1)$$

where p_j is the neighboring candidate point of p_i with a maximum distance r , and $\|\cdot\|_2$ is the Euclidean distance. If the number of neighbors $|\mathcal{N}(p_i, r)| < \tau$ with a minimum point threshold τ , the point p_i is classified as an outlier and removed.

Combining semantic constraints with density filtering enables robust point cloud clustering, facilitating the extraction of structurally continuous target regions and mitigating background interference, thereby providing a reliable basis for subsequent depth refinement and self-localization.

3.2. Depth refinement with mmWave radar assistance

Although deep learning-based monocular depth estimation generally produces reasonable dense maps, it often fails in low-texture, high-occlusion, or poorly lit regions, especially at depth edges or long

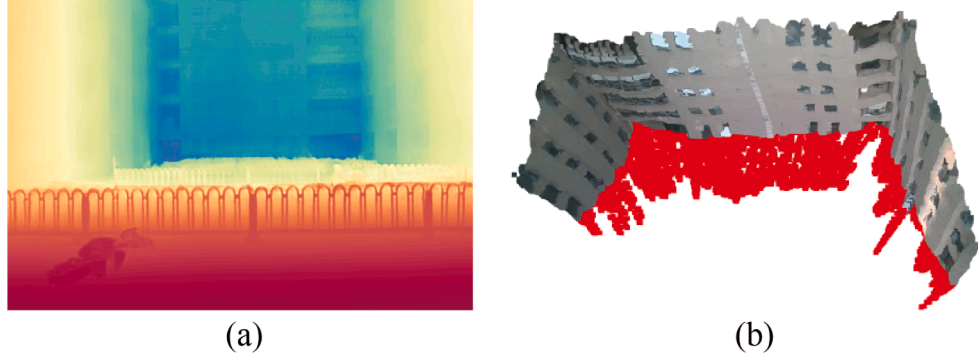


Fig. 5. Depth estimation results. (a) Predicted depth map from the Depth Anything V2 model. (b) Initial dense point cloud, with red points indicating the filtered outliers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

distances. The mmWave radar is introduced as a geometric complement, leveraging its precise distance sensing to constrain the depth structure and improve the quality of the initial point cloud.

The method corrects the reconstructed point cloud by referencing radar geometry, refining its position, orientation, and scale through staged alignment based on structural consistency. Initially, the reconstructed point cloud is aligned to the radar coordinate system using extrinsic parameters, followed by top-view planar extraction for both point clouds, which is implemented as top-view line fitting since building walls appear as straight lines in the ground projection.

3.2.1. Structural plane extraction

An improved RANSAC algorithm combined with least squares is used to enhance line fitting accuracy. Let $\mathcal{D}$ denote the point cloud dataset. In each iteration, two points $\mathbf{d}_1 : (x_1, y_1)$ and $\mathbf{d}_2 : (x_2, y_2)$ are randomly selected from $\mathcal{D}$ to construct a line $l : ax + by + c = 0$. The line parameters a , b , and c are calculated as follows:

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (2)$$

For the remaining points $\mathbf{d}_i : (x_i, y_i) \in \mathcal{D}$, the distance between $\mathbf{d}_i$ and l is given by:

$$d_i = \frac{|ax_i + by_i + c|}{\sqrt{a^2 + b^2}}. \quad (3)$$

If d_i is smaller than the distance threshold $\hat{d}_1$, then $\mathbf{d}_i$ is classified into the interior point set $\mathcal{D}_{in}$; otherwise, it is classified into the exterior point set $\mathcal{D}_{out}$, that is,

$$\begin{cases} d_i \leq \hat{d}_1, \mathbf{d}_i \in \mathcal{D}_{in} \\ d_i > \hat{d}_1, \mathbf{d}_i \in \mathcal{D}_{out} \end{cases} \quad (4)$$

In each iteration, the algorithm selects the best-fitting linear model and preserves it as the current optimal. Corresponding interior points are removed from dataset $\mathcal{D}$ to generate a new dataset for the next fitting. Each optimal model is recorded in the fitting process.

Given two lines l_1, l_2 with unit direction vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, the angle θ is calculated by:

$$\theta = \arccos \left(\frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \cdot \|\mathbf{v}_2\|} \right), \theta \in [0, 2\pi). \quad (5)$$

Lines with an angle close to zero are considered parallel. Two planes satisfying Eq. (6) are regarded as duplicates, upon which their direction, position, and corresponding point sets are merged to form a more stable geometric representation.

$$d(l_1, l_2) < \hat{d}_2, \quad (6)$$

where $d(\cdot)$ denotes the distance between the two lines in the top-view plane and $\hat{d}_2$ is a preset threshold.

A corner region detection strategy is introduced to enhance registration accuracy. For any two fitted planes, overlapping points are identified and extended along both directions. If one side contains significantly fewer points, the region is marked as a corner. This approach helps prevent structural fragmentation during rotation alignment and strengthens constraints at complex edges.

3.2.2. Establishing geometric correspondence

We project radar points onto the image plane and back-project the depth map to obtain 3D points in the camera coordinate system, which are then transformed into the radar coordinate system, allowing each radar point to be associated with a corresponding reconstruction point. Given a radar-extracted line L_{rad} with point set $\mathcal{D}_r$. Projecting $\mathcal{D}_r$ onto the image plane and reconstructing via depth map followed by coordinate transformation yields a corresponding set $\mathcal{D}_{r \rightarrow img}$. A match with a reconstructed line L_{rec} , which includes point set $\mathcal{D}_{recon}$, is confirmed when the following condition is satisfied:

$$\frac{|\mathcal{D}_{r \rightarrow img} \cap \mathcal{D}_{recon}|}{|\mathcal{D}_{r \rightarrow img}|} \geq \lambda. \quad (7)$$

The similarity threshold $\lambda \in [0, 1]$ determines whether two structural planes constitute a consistent surface-level correspondence. Such associations provide reliable geometric constraints from mmWave radar, thereby reducing structural deviations in the image-reconstructed point cloud and improving both local accuracy and overall consistency.

3.2.3. Depth refinement

After establishing structural plane correspondences, a three-stage alignment process consisting of rotation, scaling, and translation in the top view is executed on correcting geometric errors in the image-reconstructed point cloud.

For each pair of corresponding structural planes, the rotation angle $\Delta\theta$ is estimated from their direction vectors by Eq. (5). In corner areas, the intersection of the involved planes is used as a rotation anchor. When a structural plane is associated with two adjacent planes, the rotation is instead applied around its centroid.

A scaling factor is computed based on the maximum distance ratio between radar points and their reconstructed counterparts on the matched planes.

$$s = \frac{\max \|D_i^{rad} - D_j^{rad}\|}{\max \|D_i^{rec} - D_j^{rec}\|}, \quad (8)$$

where D_i^{rad}, D_j^{rad} denote radar points and D_i^{rec}, D_j^{rec} are corresponding reconstructed points, respectively. Scaling operations proceed along the principal axes determined by plane orientation, anchored at the end-point closest to the corner. When a structural plane overlaps with two

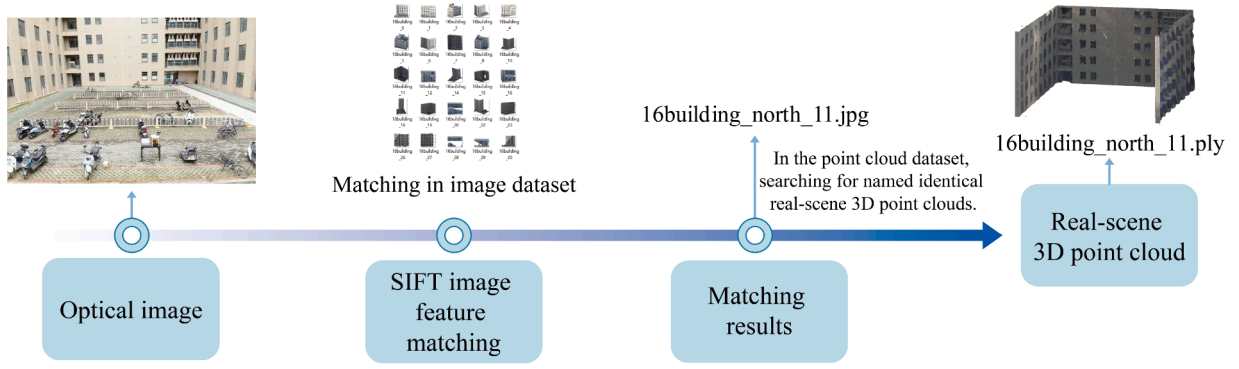


Fig. 6. SIFT matching process.

neighboring planes, joint scaling ensures local consistency, followed by a global Z-axis adjustment to correct vertical depth discrepancies.

A set of point correspondences within structural planes guides the computation of a global translation vector by averaging residuals after rotation and scaling, which subsequently translates the entire reconstructed point cloud.

By integrating radar-derived geometric constraints into orientation, scale, and translation adjustments, the reconstructed point cloud achieves improved structural integrity and spatial alignment, forming a reliable prior for subsequent self-localization.

3.3. Image matching

Although the fused point cloud generated from joint reconstruction using images and mmWave radar contains rich environmental geometry, image-based matching remains preferable during coarse alignment. The dense feature content of images in texture-rich areas facilitates rapid and precise self-localization. In contrast to the potentially noisy and unstable fused point cloud, their direct correspondence with the real-scene 3D model improves alignment reliability. Consequently, image-based coarse matching not only improves efficiency and accuracy but also offers a reliable initial guess for subsequent fine registration and pose optimization.

The SIFT algorithm is a robust local feature matching method that effectively handles variations in scale, rotation, illumination, image blur, and viewpoint. Unlike pixel-based methods, the SIFT algorithm maintains stable feature extraction even with degraded image quality or significant scene changes. Optical images are matched against a pre-constructed image database using SIFT to retrieve candidate views. Based on the image matching results, corresponding real-scene 3D model data is extracted from the point cloud dataset to estimate the approximate location within the large-scale model, as illustrated in Fig. 6.

3.4. The point cloud localization algorithm

Refined self-localization and accurate coordinate estimation are achieved by matching the depth-corrected reconstructed point cloud with the real-scene 3D data retrieved through image matching.

The centroid of the real-scene 3D point cloud is computed by:

$$\mathbf{c} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i, \quad (9)$$

where $\mathbf{c}$ represents the center of mass and $\mathbf{x}_i \in \mathbb{R}^3$ denotes the 3D points in the cloud. The point cloud is centered at the origin with its centroid offset recorded. Voxel downsampling reduces point density, with average neighborhood color assigned to the downsampled points, generating a sparse colored point cloud for accelerated processing.

This paper extracts texture keypoints based on the Hue-Saturation-Intensity (HIS) color space. Following the RGB-to-HSI conversion, the hue and saturation channels describe color type and intensity, allowing for the identification of regions with strong color characteristics, such as glass. To ensure adaptability to point density and window size, the neighborhood radius is set as $r = 5\delta_p$, with δ_p denoting the average point spacing. For each point, the mean hue difference and average saturation within its neighborhood are combined to obtain a color gradient, which highlights edges and reflective features. After normalization, points with gradient values exceeding a preset threshold are selected as texture keypoints.

The density-based spatial clustering of applications with noise (DBSCAN) is first applied to the identified keypoints, generating a initial set of candidate window regions. A secondary clustering strategy is then employed to further refine the segmentation of window areas. Clusters with abnormal bounding box dimensions are removed. Subsequently, cluster merging is guided by three types of similarity metrics: (1) Geometric size similarity, where the aspect ratios of bounding boxes are compared to determine the consistency in window dimensions; (2) Projected contour similarity, in which cluster point clouds are projected onto their principal planes via Principal Component Analysis (PCA). The aspect ratios of the resulting 2D contour acts as an indicator of shape consistency, helping to mitigate the effects of minor viewpoint deviations or partial occlusions; (3) Color similarity, where color features are converted into the Lab color space and normalized histograms are computed to measure differences in appearance, capturing consistency in window frames or glass. The three similarity metrics are fused with pre-defined weights to define the distance measure for agglomerative hierarchical clustering, producing window groupings with consistent structural and visual features. Each resulting group comprises at least two window clusters.

This paper proposes a point cloud registration method leveraging window void constraints. Here, the source point cloud refers to the reconstructed point cloud, and the target point cloud refers to the real-scene 3D model point cloud. Specifically, for each source and target point cloud group, the regions associated with texture keypoints are excluded, yielding point cloud copies with matching void structures as the geometric inputs for alignment. Let $\mathcal{T} = \{\mathbf{R}, \mathbf{T}\}$ denote the rigid transformation, where $\mathbf{R} \in \mathbb{R}^{3 \times 3}$ and $\mathbf{T} \in \mathbb{R}^3$ denote the rotation matrix and translation vector to be optimized. The dual-constraint error function $e(\mathcal{T})$ is then formulated as:

$$e(\mathcal{T}) = \alpha e_{\text{geo}}(\mathcal{T}) + \beta e_{\text{void}}(\mathcal{T}), \quad (10)$$

where α and β are weighting coefficients balancing the contribution of geometric fitting and void region alignment, respectively. $e_{\text{geo}}(\mathcal{T})$ is geometric alignment error based on the Euclidean distance between nearest point pairs, following the standard Iterative Closest Point (ICP) formulation, which can be expressed as:

$$e_{\text{geo}}(\mathcal{T}) = \sum \|\mathbf{R} \cdot \tilde{\mathbf{p}}_i + \mathbf{T} - \tilde{\mathbf{p}}'_i\|^2, \quad (11)$$

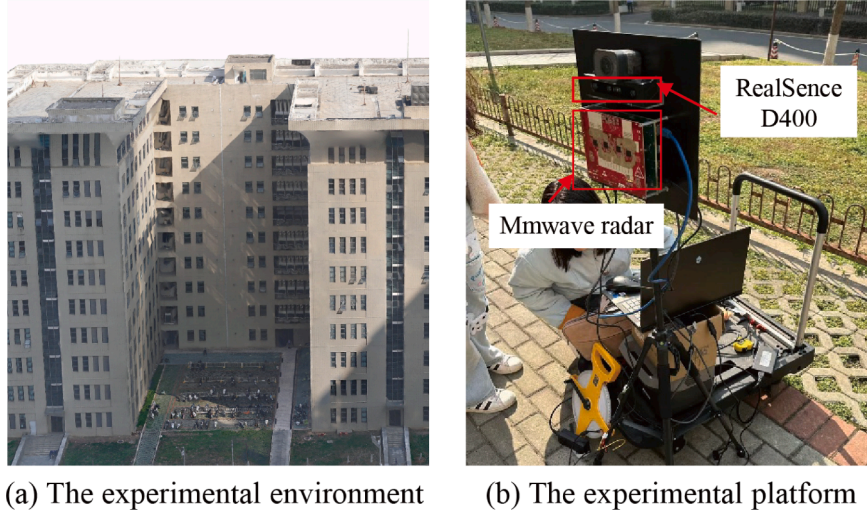


Fig. 7. Experimental setup.

where $\tilde{p}_i$ represents a point from the source point cloud and $\tilde{p}'_i$ is the corresponding nearest neighbor in the target point cloud. $e_{void}(T)$ evaluates the registration error between boundary points of the corresponding void regions as follows:

$$e_{void}(T) = \sum \left\| \mathbf{R} \cdot \tilde{b}_k + \mathbf{T} - \tilde{b}'_k \right\|^2, \quad (12)$$

where $\tilde{b}_k$ is the boundary point set of the void area in the source point cloud and $\tilde{b}'_k$ represents the corresponding boundary point in the target point cloud. A RANSAC algorithm is introduced by randomly selecting initial boundary correspondences to estimate the rigid transformation. All resulting candidates are evaluated using a combined alignment error, retaining the one with the lowest score. Adding the corresponding translation vector to the subtracted centroid offset yields the final position in the real-scene 3D model, ensuring accurate alignment and self-localization.

The proposed semantic-guided fine registration exploits the structural regularity of window geometry to improve alignment between UAV observations and the 3D model. By incorporating semantic cues as geometric constraints, the method achieves more accurate registration and increased robustness under noisy or incomplete sensing. This contribution demonstrates the value of semantic information in ensuring reliable self-localization in urban environments, underscoring its broader relevance of semantic information in supporting reliable UAV self-localization in complex urban environments [32].

4. Experiments and results

An autonomous UAV self-localization system was implemented and validated through experiments conducted in the U-shaped open-air parking area of the campus experimental teaching building (Fig. 7(a)). The system coordinates an Intel RealSense D400 camera with a Texas Instruments four-chip mmWave cascaded radar (Fig. 7(b)). The cascaded radar supports up to 12 transmit antennas and 16 receive antennas, enabling continuous transmission and reception of frequency-modulated radio waves. To accommodate detection in dense building environments, the maximum detection range of the radar is set to 45 m. Detailed parameters are listed in Table 2.

4.1. Sensor calibration

4.1.1. Radar and camera individual calibration

This paper utilizes a four-chip cascaded mmWave radar. Since array systems formed by multiple cascaded radars can exhibit channel amplitude and phase inconsistencies as well as mutual coupling effects among

Table 2

Parameter and their values of mmWave radar.

Parameter	Value
Rx	16
Tx	12
Start frequency (GHz)	77
Frequency slope (MHz/ μ s)	78.9
Samples per chip	512
ADC sample frequency (ksps)	24,000

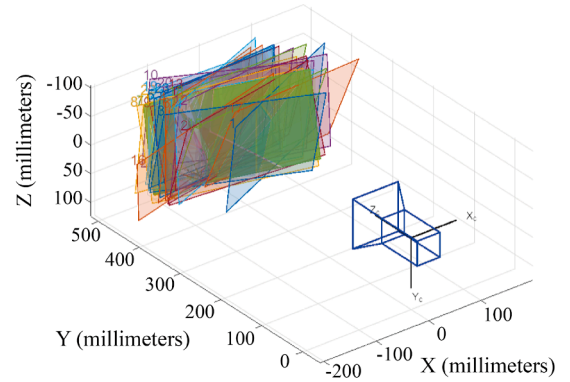


Fig. 8. Zhang Zhengyou's calibration method, with the coordinate system centered on camera.

antenna elements, a channel calibration method is applied to compensate deviations, improving the consistency and accuracy of the array data.

Using Zhang Zhengyou's calibration method [33] implemented in MATLAB's Camera Calibrator, the camera intrinsic parameters are estimated from Multiple images of a 6×9 checkerboard captured at varying angles and distances. Extracted corner points, combined with the known physical dimensions, enable computation of the pixel-to-world mapping and subsequent nonlinear optimization to obtain the intrinsic matrix, as depicted in Fig. 8.

4.1.2. Radar and camera joint calibration

In 3D space, a physical target is represented with different coordinates in different sensor coordinate systems. The goal of sensor fusion calibration is to unify these heterogeneous coordinates within a com-

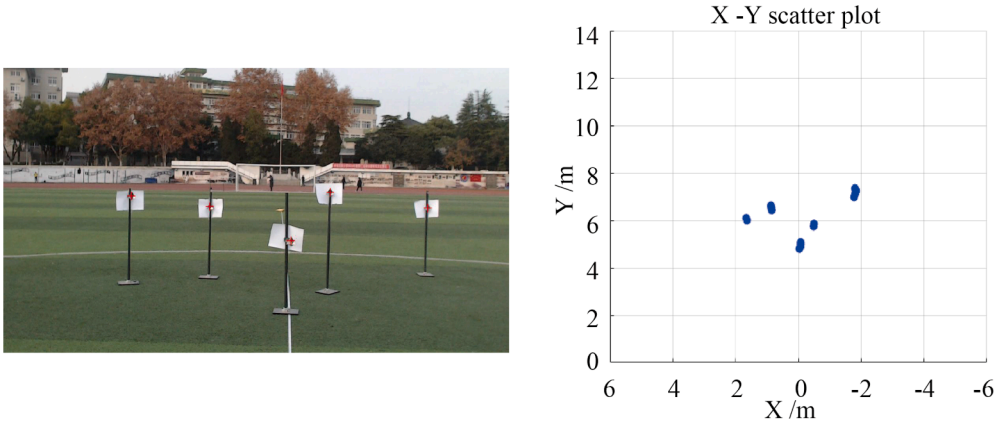


Fig. 9. Radar reflector points and corresponding reflectors in the camera frame.

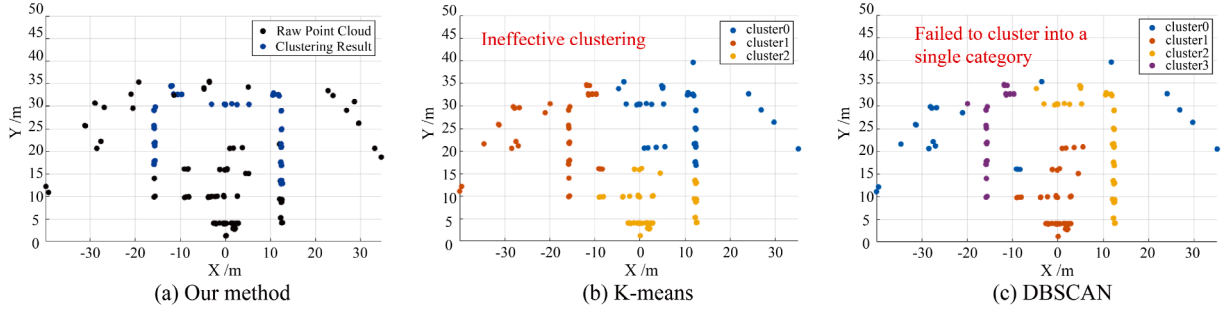


Fig. 10. 2D view of clustering results on mmWave radar point clouds. (a) Result of the proposed clustering method. (b) K-means clustering. (c) DBSCAN clustering.

mon reference frame. Assuming there is a point in real world denoted as $\mathbf{v} : (x_v, y_v)^T$ in the pixel coordinate system and $\mathbf{w} : (x_w, y_w, z_w)^T$ in the radar coordinate system. The pair $(\mathbf{v}, \mathbf{w})$ defines a radar-image correspondence. This paper adopts the radar coordinate system as the world coordinate system. The transformation between the radar and pixel coordinate systems is given by:

$$Z_c \begin{bmatrix} x_v \\ y_v \\ 1 \end{bmatrix} = \mathbf{M} \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix}, \quad (13)$$

where $\mathbf{R}$ and $\mathbf{T}$ denote the rotation and translation between the two sensors, respectively. In addition, $\mathbf{M}$ is the intrinsic parameter matrix of the camera and Z_c denotes the point's z-value in camera coordinate. As shown in Fig. 9, multiple radar-image point pairs are obtained using a corner reflector and filtered via the RANSAC algorithm to remove outliers. The resulting correspondences serve as input to the solvePnP function in OpenCV for estimating the camera's extrinsic parameters. Using Eq. (13), each radar coordinate (x_w, y_w, z_w) is projected into the image plane to obtain its corresponding pixel coordinate (x'_v, y'_v) . The average calibration error is then defined as:

$$e_{\text{proj}} = \frac{1}{N_c} \sum_{i=1}^{N_c} \sqrt{(x_{v,i} - x'_{v,i})^2 + (y_{v,i} - y'_{v,i})^2}, \quad (14)$$

where N_c is the number of calibration correspondences.

Calibration results show a mean reprojection error of approximately 0.43 pixels, which is within one pixel on average, indicating that the calibration precision is sufficient to meet the requirements for reliable radar-camera fusion [34].

4.2. Radar data processing

The mmWave radar is a robust sensing system capable of reliably detecting target positions and velocities under various challenging con-

ditions. The raw radar data collected during the experiment undergo a processing pipeline including ADC calibration, range FFT, Doppler FFT, constant false alarm rate (CFAR) detection, and direction of arrival (DoA) estimation, producing point clouds with 3D spatial coordinates (x, y, z) as well as velocity and reflectivity attributes. The spatial points capture the geometric profile, structural characteristics, and motion state of the detected targets. Additionally, Doppler information is exploited to estimate the motion of detected points, facilitating the extraction of static target point clouds.

4.3. Experimental procedure

The experiment is conducted in a U-shaped open-air parking area near the experimental teaching building on campus, as illustrated in Fig. 7(a). The perception system is mounted on a static platform, capturing high-resolution optical images and sparse mmWave point clouds. Through processing and fusing these multimodal data, accurate self-localization within the real-scene 3D model is accomplished.

We first acquire a real-time environmental representation through optical images and generate an initial dense point cloud using depth estimation combined with semantic segmentation. Before integrating radar data, we perform a semantic-guided clustering method to the mmWave radar point clouds. Fig. 10 shows the clustering results of our method in comparison with K-means and DBSCAN algorithm. Through several experiments, the optimal DBSCAN parameter are set to $\text{Eps}=6$ and $\text{MinPts}=8$. Fig. 10(a) demonstrates the clustering results of the proposed algorithm. Despite differences in camera and radar viewpoints cause some target points to be missed during clustering, the method successfully achieves reliable clustering and effectively extracts the structurally continuous target region. As depicted in Fig. 10(b), the K-means algorithm fails to yield meaningful clustering results. The DBSCAN algorithm removes most outliers but produces fragmented clusters that lack coherence, as shown in Fig. 10(c). In contrast, the proposed algorithm

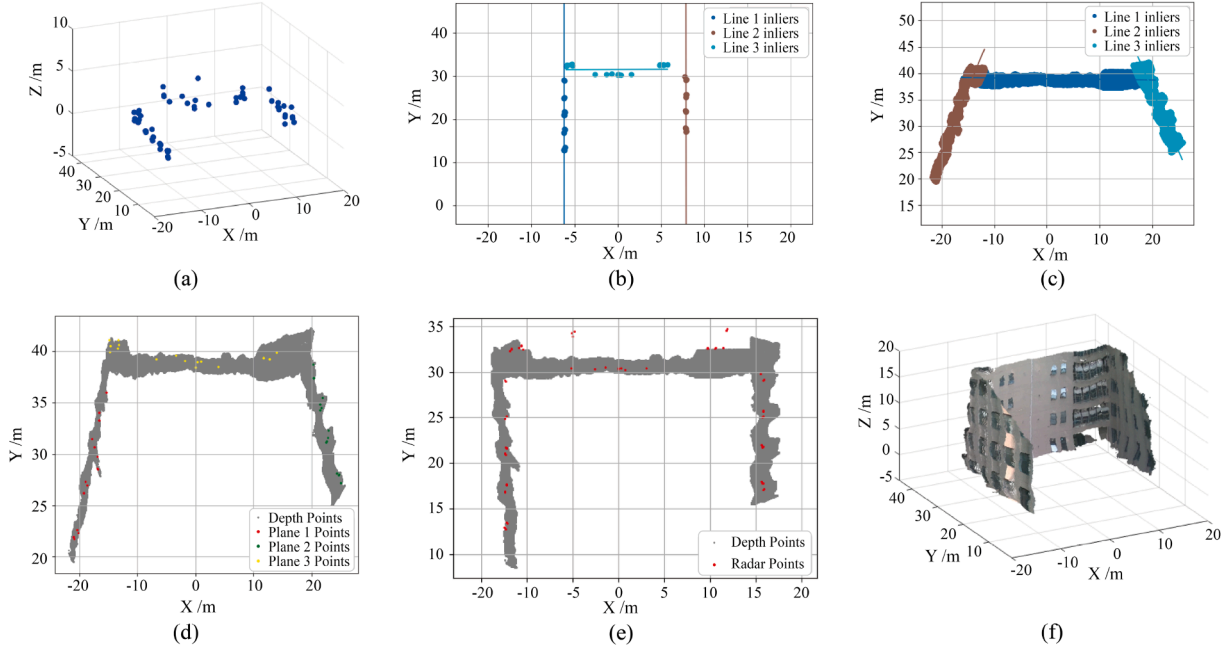


Fig. 11. Visualization of radar-image fusion processing. (a) Semantic-guided radar clustering. (b)–(c) Structural surfaces from radar and reconstruction. (d) Geometric correspondences via depth projection. (e) 2D and (f) 3D depth correction visualizations.

achieves more effective clustering of the target point cloud, demonstrating superior accuracy and stability.

The mmWave radar point clouds are incorporated to complement geometry and correct errors, as presented in Fig. 11, which summarizes the full correction pipeline. Fig. 11(a) displays the result of radar point cloud clustering guided by semantic information. This approach effectively removes non-structural noise, achieves an extraction accuracy exceeding 90 %, and preserves key geometric structures, thereby providing a highly reliable data foundation for subsequent point cloud registration. Fig. 11(b) and (c) show the extracted structural surfaces from radar and reconstructed point clouds, respectively, highlighting the geometric consistency across modalities. Fig. 11(d) offers the geometric correspondences established via depth projection, where the colored points represent radar reflections mapped to their corresponding positions in the reconstructed point cloud, and the gray points denote the complete reconstructed cloud. Fig. 11(e) shows the 2D visualization of depth correction results, with red points representing radar points and gray points indicating the corrected reconstruction point cloud. Fig. 11(f) provides the corresponding 3D visualization. Collectively, Fig. 11 demonstrates that incorporating radar geometry enhances both the structural reliability and accuracy of the reconstructed scene.

Following depth correction, a coarse self-localization is performed by retrieving the corresponding region from the real-scene 3D model via image matching. As depicted in Fig. 12, the local geometric distribution of windows is incorporated to introduce an auxiliary registration method, enabling precise alignment with the high-resolution real-scene 3D model. Fig. 12(a) and (b) show the detected window regions and their spatial distributions in the reconstructed point cloud and the 3D model, respectively. Different colors indicate distinct window clusters, each serving as a local semantic anchor to improve the accuracy and robustness of point cloud alignment. Fig. 12(c) presents a top-view 2D projection of the final registration, while Fig. 12(d) displays the corresponding 3D registration outcome. The final self-localization result produced by the proposed method is (537450.95 m, 3375091.12 m, 1.34 m).

To validate the accuracy of the proposed algorithm, the system is manually shifted by one meter in the left, right, forward and backward directions, with data collected at each position. The system coordinates are estimated using the proposed algorithm. Simultaneously, a laser

Table 3

The drone positioning errors.

	1	2	3	4	5	6	7	8	9	10
Error (m)	0.288	0.347	0.205	0.275	0.346	0.327	0.389	0.321	0.186	0.34

rangefinder records the distance from the detection point to surrounding wall, providing ground-truth coordinates for comparison. Fig. 13 illustrates comparison between estimated and reference coordinates in the X and Y axes, along with the corresponding positioning errors. The drone positioning error is calculated by:

$$e_{\text{loc}} = \sqrt{(\Delta x)^2 + (\Delta y)^2}, \quad (15)$$

where Δx and Δy represent the differences between the estimated and reference UAV positions in the X and Y axes, respectively.

The positioning error data for the ten test location are listed in Table 3. Fig. 14 shows the cumulative distribution function (CDF) of the positioning error with the horizontal axis indicating the error magnitude and the vertical axis representing the corresponding CDF values. As observed in the figure, the distance measurement error fluctuates within a certain range, achieving an accuracy of up to 0.4 m. The results demonstrate strong consistency in global structural alignment and local boundary matching, confirming reliable self-localization performance in static and structurally complex environments.

4.4. Results analysis

Successful clustering of radar point clouds is a critical step for subsequent processing. This paper proposes a semantic-guided clustering method to precisely extract target regions. Evaluations involved three experiments in the test scenario are summarized in Table 4, showing that the proposed method attains over 90 % precision, surpassing both K-means and DBSCAN algorithms.

After validating the effectiveness of the proposed clustering method, we further evaluated the overall localization accuracy of the system. This step is critical to demonstrate how semantic-guided radar correction contributes beyond noise suppression, by directly influencing the

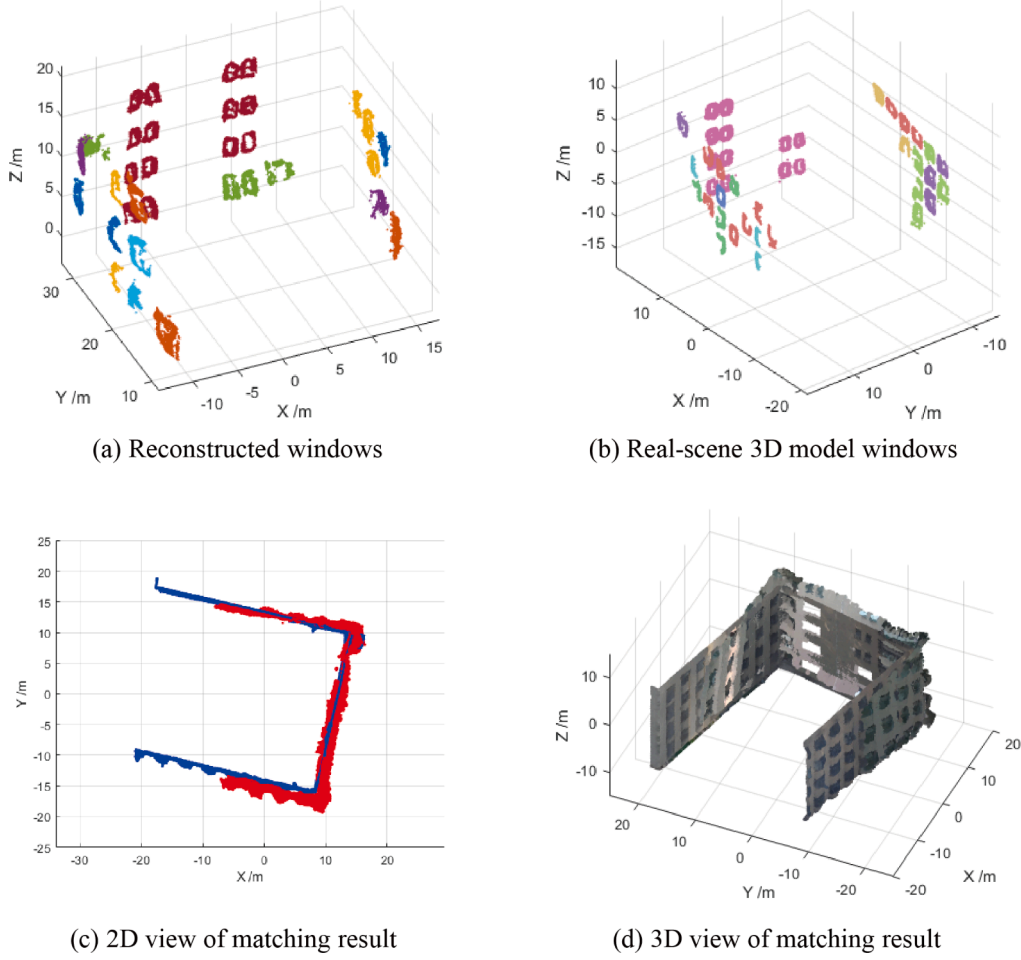


Fig. 12. Window spatial distribution aiding high-precision self-localization.

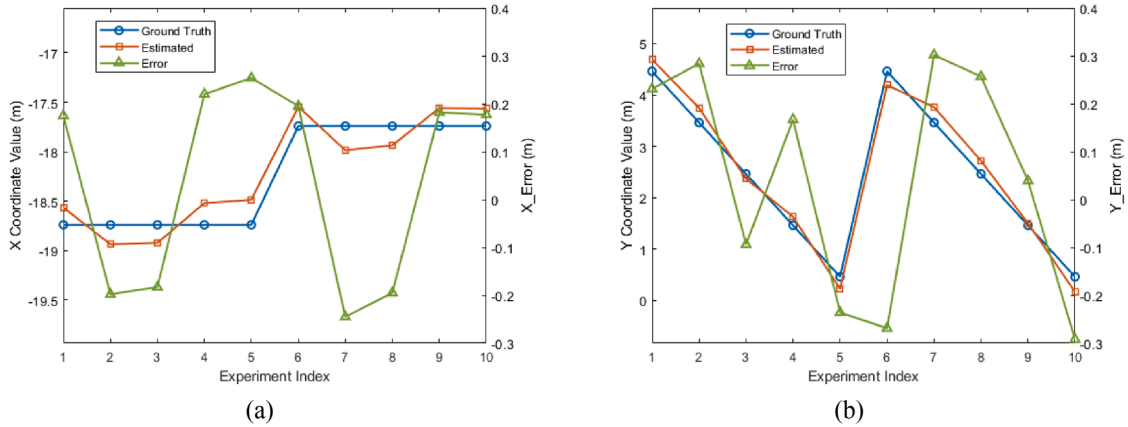


Fig. 13. Position estimation results at 10 different locations.

reliability of global alignment with the reference 3D model. To ensure a fair and transparent evaluation, two baseline methods were designed for comparison under identical conditions.

Vision-only: This baseline generates point clouds solely from monocular depth estimation while keeping the same registration pipeline as in our method. As discussed in Section 3.2, depth estimation from monocular vision inherently suffers from scale ambiguity and prediction bias, which severely limits its reliability for UAV localiza-

tion. This baseline thus directly illustrates the improvement achieved by radar correction.

Vision + Radar without fine registration: In this baseline, both modalities are fused while the semantic-guided refinement step is omitted. This configuration allows us to isolate the specific contribution of fine registration to the overall alignment accuracy.

Here, Depth Root Mean Square (DRMS) and Mean Radial Spherical Error (MRSE) are introduced as evaluation metrics to assess

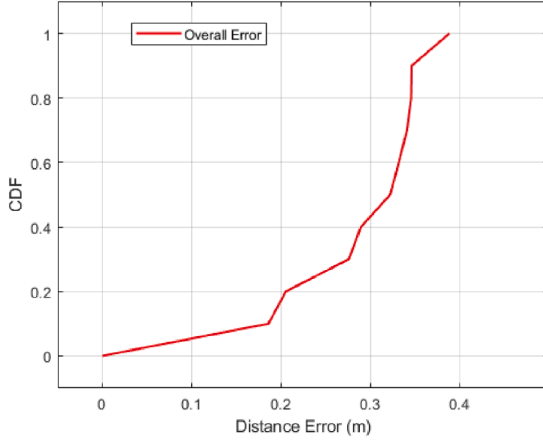


Fig. 14. The positioning error diagram.

Table 4
Precision (%) of different point cloud clustering methods.

Experiment ID	Ours	K-means	DBSCAN
1	94.9	54.9	53.4
2	90.4	53.2	53.1
3	92.3	52.9	56.1

self-localization accuracy:

$$DRMS = \sqrt{\frac{1}{N_o} \sum_{i=1}^{N_o} \Delta x_i^2}, \quad (16)$$

$$MRSE = \sqrt{\frac{1}{N_o} \sum_{i=1}^{N_o} (\Delta x_i^2 + \Delta y_i^2 + \Delta z_i^2)}, \quad (17)$$

where N_o is the number of observations in each experiment and Δx_i , Δy_i , Δz_i denote the localization errors along the X, Y, Z-axes in the i th observation. Three groups of experiments were conducted in the selected environment, each consisting of 10 data samples.

The comparative results are summarized in Fig. 15. To highlight the performance differences among the methods more clearly, the MRSE, represented by the red line, and the standard deviation (Std), represented by the blue bars, are reported in Fig. 15(a). This figure provides a comparative chart of localization performance, offering a comprehensive evaluation of each method's accuracy and stability. Specifically,

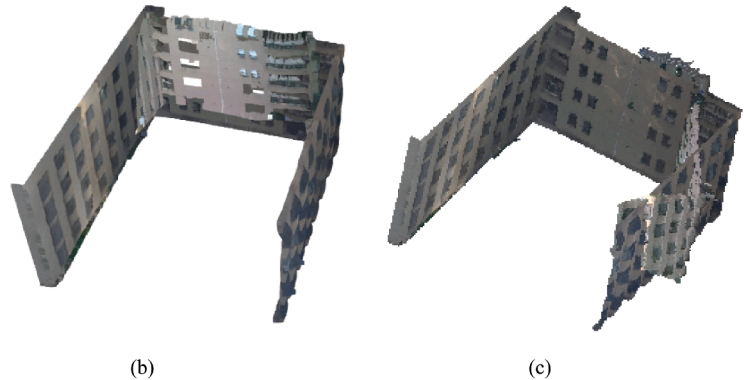
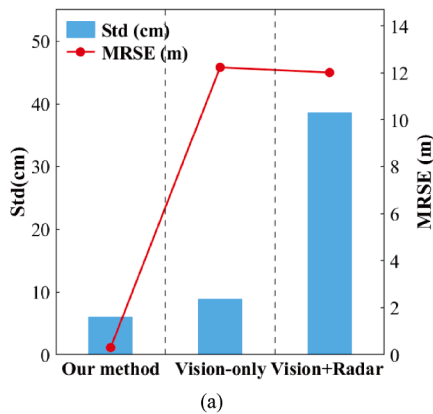


Fig. 15. Comparison of localization error and registration results under different methods. (a) Localization error ($MRSE \pm Std$) under different methods. (b) Registration result of Vision-only baseline. (c) Registration result of Radar + Vision without fine registration baseline.

MRSE reflects the average positioning accuracy, while Std indicates the robustness of the results across multiple trials.

As shown in Fig. 15(a), the MRSE values of the Vision-only, Radar + Vision (without fine registration), and proposed methods are 12.23 m, 12.01 m, and 0.302 m, respectively. The corresponding Std values are 0.386 m, 0.088 m, and 0.06 m, respectively. These results clearly indicate that the proposed method achieves both the highest accuracy and the strongest stability among all tested configurations. Representative registration outcomes for the two baselines are further illustrated in Fig. 15(b) (c), where significant misalignments can be observed.

As expected, the Vision-only baseline suffers from large bias and frequent failures due to inherent depth estimation errors, with localization errors often exceeding several meters. Although its standard deviation remains relatively small, suggesting a stable but biased pattern, the overall registration is limited to coarse alignment. This further motivates the integration of mmWave radar for geometric correction. The Radar + Vision baseline without semantic-guided refinement achieves slightly reduces the mean error by incorporating radar depth cues, but its variance increases and misalignments occur more frequently. This degradation mainly stems from the absence of semantic constraints, which prevents the registration process from resolving geometric ambiguities and thus leads to unstable alignment outcomes. In contrast, the full radar-vision fusion pipeline consistently achieves a mean localization error within 0.4 m, as further quantified by the DRMS and MRSE metrics shown in Fig. 16. Furthermore, the qualitative registration results in Fig. 12(c) (d) demonstrate precise alignment with the reference model, in sharp contrast to the misaligned outcomes observed in Fig. 15(b) (c).

Note that we did not include a radar-vision fusion baseline without semantic clustering, since the absence of semantic constraints makes it challenging to establish reliable correspondences between radar and visual point clouds, often resulting in unstable or failed alignment. Instead, we consider (i) a vision-only baseline to highlight the corrective benefit of radar, and (ii) a radar-vision fusion baseline without semantic refinement to explicitly evaluate the contribution of semantic guidance.

According to the results in Fig. 16, the average DRMS and MRSE in this experimental scenario reach 0.2086 m and 0.3242 m, respectively, supporting the reliability and practical feasibility of the proposed method.

The proposed method enables autonomous systems to accurately obtain positional information within real-world environments. Compared to most existing methods, this method adopts a simpler sensor setup and lower system complexity. Moreover, the dataset constructed from a real-scene model allows direct application for self-localization without requiring any pre-training. A comparative analysis of key parameters between the proposed method and several representative self-localization approaches is presented in Table 5.

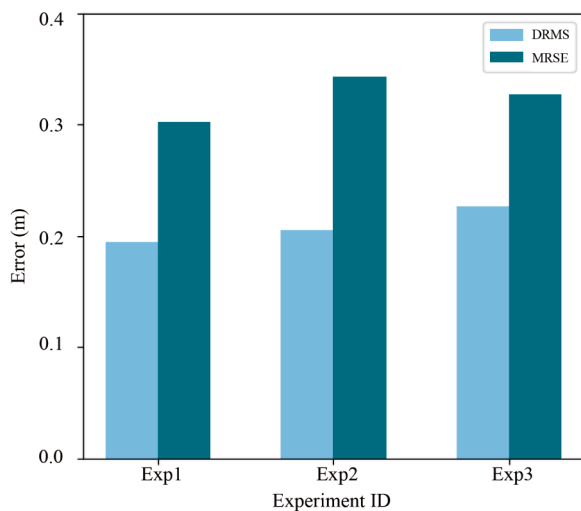


Fig. 16. Self-localization errors across multiple experimental groups.

Table 5
Comparison with other approaches.

	Ours	Ref. [35]	Ref. [36]
Data preprocessing	No	Yes	Yes
Number of sensors	2	4	3
Positioning accuracy	0.4 m	1.66 m	0.5 m

5. Conclusion

This paper addresses the problem of self-localization for UAVs under GNSS-denied conditions by proposing a 3D mapping and localization method that integrates mmWave radar with visual sensing. An initial dense point cloud is constructed through image-based depth estimation and semantic segmentation, while geometric errors are corrected using mmWave radar point clouds, enhancing overall mapping accuracy. To enhance structural reliability, a semantic-guided clustering and structural extraction method is introduced to filter radar noise and identify key architectural features such as wall surfaces. During positioning, the spatial distribution of window cavities is leveraged as auxiliary constraints to achieve precise alignment between local point clouds and the real-scene 3D model. Experimental results demonstrate that the proposed method delivers sub-meter self-localization accuracy in typical building environments, while maintaining low system complexity and deployment cost, thereby offering a promising solution for UAV self-localization in complex urban scenarios.

Nevertheless, the study has several limitations. **First**, the experiments were conducted in a localized campus environment, which may restrict generalization to larger and more diverse urban areas. **Second**, the system has not yet been deployed on UAVs, and real-time embedded implementation remains challenging due to payload, power, and processing constraints. **Third**, the current framework is tailored for outdoor façades and does not yet address indoor environments, where multipath effects and weaker radar returns require specialized treatment. Future work will therefore focus on extending the framework to larger-scale and more diverse urban areas, as well as exploring adaptability to different architectural styles. Additional efforts will be devoted to improving detection inside buildings through radar denoising and SLAM-based integration, developing lightweight real-time algorithms suitable for UAV platforms, and incorporating advanced fusion models such as transformers or graph neural networks to further enhance accuracy. These directions will broaden the scalability and robustness of the system, making it a more practical solution for UAV self-localization in complex and dynamic urban environments.

CRedit authorship contribution statement

Yongqiang Cui: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization; **Yiyang Zhang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Di Bai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition; **Yi Diao:** Writing – review & editing, Investigation, Data curation; **Yulei Wang:** Writing – review & editing, Supervision.

Data availability

All of the required data are available in the manuscript. Raw data can be obtained upon reasonable request via Email.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] S. Jiang, W. Jiang, L. Wang, Unmanned aerial vehicle-based photogrammetric 3D mapping: a survey of techniques, applications, and challenges, *IEEE Geosci. Remote Sens. Mag.* 10 (2) (2022) 135–171. <https://doi.org/10.1109/MGRS.2021.3122248>
- [2] L. Yang, J. Fan, Y. Liu, E. Li, J. Peng, Z. Liang, A review on state-of-the-art power line inspection techniques, *IEEE Trans. Instrum. Meas.* 69 (12) (2020) 9350–9365. <https://doi.org/10.1109/TIM.2020.3031194>
- [3] M. Bakirci, Vehicular mobility monitoring using remote sensing and deep learning on a UAV-based mobile computing platform, *Measurement* 244 (2025) 116579. <https://doi.org/10.1016/j.measurement.2024.116579>
- [4] Y. Qin, M.A. Kishk, M.-S. Alouini, Stochastic geometry-based trajectory design for multi-purpose UAVs: package and data delivery, *IEEE Trans. Veh. Technol.* 73 (3) (2024) 4136–4150. <https://doi.org/10.1109/TVT.2023.3323682>
- [5] M. Lyu, Y. Zhao, C. Huang, H. Huang, Unmanned aerial vehicles for search and rescue: a survey, *Remote Sens.* 15 (13) (2023). <https://doi.org/10.3390/rs15133266>
- [6] N. Zhu, J. Marais, D. Bétaille, M. Berbineau, GNSS position integrity in urban environments: a review of literature, *IEEE Trans. Intell. Transp. Syst.* 19 (9) (2018) 2762–2778. <https://doi.org/10.1109/TITS.2017.2766768>
- [7] Y. Wang, L. Feng, Y. Liu, Z. Li, Stochastic geometry analysis for information integration and communication in cellular and D2D-based heterogeneous IoT, *Comput. Netw.* 261 (2025) 111126. <https://doi.org/10.1016/j.comnet.2025.111126>
- [8] Y. Wang, Y. Liu, Y. Fu, Y. Qin, Z. Li, 3D stochastic geometry model for aerial vehicle-relayed ground-air-satellite connectivity, 2025, 2503.16202
- [9] E. Rublee, V. Rabaud, K. Konolige, G. Bradski, ORB: an efficient alternative to SIFT or SURF, in: 2011 International Conference on Computer Vision, 2011, pp. 2564–2571. <https://doi.org/10.1109/ICCV.2011.6126544>
- [10] D.G. Lowe, Distinctive image features from scale-invariant keypoints, *Int. J. Comput. Vision* 60 (2) (2004) 91–110. <https://doi.org/10.1023/B:VISI.0000029664.99615.94>
- [11] D. Durmuş, U. Gündükbay, Ö. Ulusoy, Learning visual similarity for image retrieval with global descriptors and capsule networks, *Multimed. Tools Appl.* 83 (7) (2024) 20243–20263. <https://doi.org/10.1007/s11042-023-16164-5>
- [12] A. Couturier, M.A. Akhloufi, A review on absolute visual localization for UAV, *Rob. Auton. Syst.* 135 (2021) 103666. <https://doi.org/10.1016/j.robot.2020.103666>
- [13] X. Qiu, S. Liao, D. Yang, Y. Li, S. Wang, High-precision visual geo-localization of UAV based on hierarchical localization, *Expert Syst. Appl.* 267 (2025) 126064. <https://doi.org/10.1016/j.eswa.2024.126064>
- [14] G. Liu, C. Li, S. Zhang, Y. Yuan, VL-MFL: UAV visual localization based on multi-source image feature learning, *IEEE Trans. Geosci. Remote Sens.* 62 (2024) 1–12. <https://doi.org/10.1109/TGRS.2024.3383509>

- [15] J. Qian, K. Chen, Q. Chen, Y. Yang, J. Zhang, S. Chen, Robust visual-lidar simultaneous localization and mapping system for UAV, *IEEE Geosci. Remote Sens. Lett.* 19 (2022) 1–5. <https://doi.org/10.1109/LGRS.2021.3099166>
- [16] P. Petráček, V. Krátký, M. Saska, Dronument: system for reliable deployment of micro aerial vehicles in dark areas of large historical monuments, *IEEE Rob. Autom. Lett.* 5 (2) (2020) 2078–2085. <https://doi.org/10.1109/LRA.2020.2969935>
- [17] Y. Xu, Z. Chen, C. Deng, S. Wang, J. Wang, LCDL: toward dynamic localization for autonomous landing of unmanned aerial vehicle based on LiDAR–camera fusion, *IEEE Sens. J.* 24 (16) (2024) 26407–26415. <https://doi.org/10.1109/JSEN.2024.3424218>
- [18] B. Yang, E. Yang, L. Yu, C. Niu, Ultrasonic- and IMU-based high-precision UAV localization for the low-cost autonomous inspection in oil and gas pressure vessels, *IEEE Trans. Ind. Inf.* 19 (10) (2023) 10523–10534. <https://doi.org/10.1109/TII.2023.3240874>
- [19] C. Xiong, W. Lu, H. Xiong, H. Ding, Q. He, D. Zhao, J. Wan, F. Xing, Z. You, Onboard cooperative relative positioning system for micro-UAV swarm based on UWB/vision/INS fusion through distributed graph optimization, *Measurement* 234 (2024) 114897. <https://doi.org/10.1016/j.measurement.2024.114897>
- [20] S. Kuutti, S. Fallah, K. Katsaros, M. Dianati, F. McCullough, A. Mouzakitis, A survey of the state-of-the-art localization techniques and their potentials for autonomous vehicle applications, *IEEE Internet Things J.* 5 (2) (2018) 829–846. <https://doi.org/10.1109/JIOT.2018.2812300>
- [21] J. Liu, Z. Gao, Y. Li, S. Lv, J. Liu, C. Yang, Ranging offset calibration and moving average filter enhanced reliable UWB positioning in classic user environments, *Remote Sens.* 16 (14) (2024). <https://doi.org/10.3390/rs16142511>
- [22] Q. Zeng, Y. Jin, H. Yu, X. You, A UAV localization system based on double UWB tags and IMU for landing platform, *IEEE Sens. J.* 23 (9) (2023) 10100–10108. <https://doi.org/10.1109/JSEN.2023.3260311>
- [23] G. Elghazaly, R. Frank, S. Harvey, S. Safko, High-definition maps: comprehensive survey, challenges, and future perspectives, *IEEE Open J. Intell. Transp. Syst.* 4 (2023) 527–550. <https://doi.org/10.1109/OJITS.2023.3295502>
- [24] Z. Gu, S. Cheng, C. Wang, R. Wang, Y. Zhao, Robust visual localization system with HD map based on joint probabilistic data association, *IEEE Rob. Autom. Lett.* 9 (11) (2024) 9415–9422. <https://doi.org/10.1109/LRA.2024.3457375>
- [25] N. Li, J. Chen, S. Guo, F. Yu, Y. Yao, G. Cui, X. Yang, Building layout tomographic imaging with improved delay estimation algorithm, *IEEE Antennas Wireless Propag. Lett.* 22 (5) (2023) 1079–1083. <https://doi.org/10.1109/LAWP.2022.3232769>
- [26] C.X. Lu, M.R.U. Saputra, P. Zhao, Y. Almalioglu, P.P.B. de Gusmao, C. Chen, K. Sun, N. Trigoni, A. Markham, MilliEgo: single-chip mmWave radar aided egomotion estimation via deep sensor fusion, in: *Proceedings of the 18th Conference on Embedded Networked Sensor Systems, SenSys '20*, Association for Computing Machinery, New York, NY, USA, 2020, p. 109–122. <https://doi.org/10.1145/3384419.3430776>
- [27] Y. Chen, Y. Zhuang, B. Wang, J. Huai, 4D radarPR: context-aware 4D radar place recognition in harsh scenarios, *ISPRS J. Photogramm. Remote Sens.* 221 (2025) 210–223. <https://doi.org/10.1016/j.isprsjprs.2025.01.033>
- [28] A. Kirillov, R. Girshick, K. He, P. Dollár, Panoptic feature pyramid networks, in: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 6392–6401. <https://doi.org/10.1109/CVPR.2019.00656>
- [29] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, H. Zhao, Depth anything v2, 2024, 2406.09414
- [30] X. Li, H. Yuan, W. Li, H. Ding, S. Wu, W. Zhang, Y. Li, K. Chen, C.C. Loy, OMG-Seg: Is one model good enough for all segmentation?, *2024 IEEE CVF Conf. Comput. Vision Pattern Recognit. (CVPR)* (2024) 27948–27959.
- [31] B. Cheng, I. Misra, A.G. Schwing, A. Kirillov, R. Girdhar, Masked-attention mask transformer for universal image segmentation, in: *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 1280–1289. <https://doi.org/10.1109/CVPR52688.2022.00135>
- [32] K. Marco, S. Goyal, R. Kaur, A review of underwater wireless sensor networks deployment techniques and challenges, *Babylonian J. Netw.* 2024 (2024) 111–121. <https://doi.org/10.58496/BJN/2024/012>
- [33] Z. Zhang, A flexible new technique for camera calibration, *IEEE Trans. Pattern Anal. Mach. Intell.* 22 (11) (2000) 1330–1334. <https://doi.org/10.1109/34.888718>
- [34] T. Kavitha, M. Venkatesan, S. Gopalakrishnan, S.R. Chand, M. Gopianand, S. Abirami, Underwater wireless sensors increase routing performance using impact efficient localization-based routing protocols, *Babylonian J. Netw.* 2024 (2024) 69–77. <https://doi.org/10.58496/BJN/2024/008>
- [35] W. Jiang, Y. Yu, K. Zong, B. Cai, C. Rizos, J. Wang, D. Liu, W. Shangguan, A seamless train positioning system using a lidar-aided hybrid integration methodology, *IEEE Trans. Veh. Technol.* 70 (7) (2021) 6371–6384. <https://doi.org/10.1109/TVT.2021.3080393>
- [36] Z. Lan, J. Wang, Z. Shen, Z. Fang, Highly robust and accurate multi-sensor fusion localization system for complex and challenging scenarios, *Measurement* 235 (2024) 114851. <https://doi.org/10.1016/j.measurement.2024.114851>